

IMMUNITY AND ITS TYPES (INNATE, HUMORAL & CELL-MEDIATED)

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Lecturer

DIKIOHS, DUHS.

LEARNING OBJECTIVES

By the end of this lecture students should be able to:

- Define immunity & its types i.e innate immunity, cell mediated immunity & humoral immunity.
- Differentiate between active immunity & passive immunity.
- Differentiate between cell mediated immunity & humoral immunity.
- Describe the role of T & B lymphocytes in cell mediated & humoral immunity.
- Identify the key cells involved in cell mediated immunity & humoral immunity.
- Compare the mechanisms of cell mediated immunity & humoral immunity.
- Define compliment system & its classical pathway.
- Understand how both immune systems work together to defend the body against infections.

IMMUNITY

Immunity is defined as:

The capacity of the body to resist pathogenic agents.

It is the ability of body to resist the entry of different types of foreign bodies like bacteria, virus, toxic substances etc.

TYPES OF IMMUNITY

There are two types of immunity:

1. Innate immunity
2. Acquired immunity

INNATE IMMUNITY OR NON-SPECIFIC IMMUNITY

Innate immunity also called *natural* or *non-specific* immunity is present by birth and is defined as:

the inborn capacity of body to resist pathogens.

If any organism enter the body, innate immunity eliminates it before the development of any disease. This type of immunity represents the first line of defence against any type of pathogens.

STRUCTURES & MECHANISM OF INNATE IMMUNITY

- **SKIN:** Skin provides resistance to invasion by organism. The keratinized stratum corneum of epidermis protects the skin against toxic chemicals. The *beta defensins* and *lysozyme* secreted in the skin are anti-microbial peptides that destroy bacteria.

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- **GASTROINTESTINAL TRACT:** Enzymes in digestive juices and the acid secretions of stomach destroy the toxic substances and organisms entering the digestive tract through food. Lysozymes present in saliva also destroys bacteria.
- **RESPIRATORY SYSTEM:** Moist surface of nasal mucosa captures the foreign particles entering the respiratory tract. *Defensins* and *cathellicidins* in epithelial cells of air passage are anti-microbial peptides that destroy the micro-organisms. Neutrophils, lymphocytes, macrophages and natural killer cells present in lungs act against bacteria & virus.
- **UROGENITAL SYSTEM:** Acidity in the urine and vaginal fluid destroys bacteria.

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- **PHAGOCYTIC CELLS:** Neutrophils, monocytes and macrophages ingest and destroy the micro-organisms & foreign bodies by *phagocytosis*.
- **INTERFERONS & COMPLIMENT PROTEINS:** Interferons inhibit multiplication of viruses, parasites & cancer cells. Compliment proteins are a system of 20 proteins that accelerate the destruction of micro-organisms.

COMPONENTS OF INNATE IMMUNITY



Eosinophil



Epithelial barrier



Mast cell



Basophil



NK cell



Neutrophil



Dendritic cell



Complement



Macrophage

ACQUIRED IMMUNITY OR SPECIFIC IMMUNITY

Acquired immunity also known as specific immunity is the resistance developed in the body against any specific foreign body like bacteria, viruses, toxins, vaccines etc.

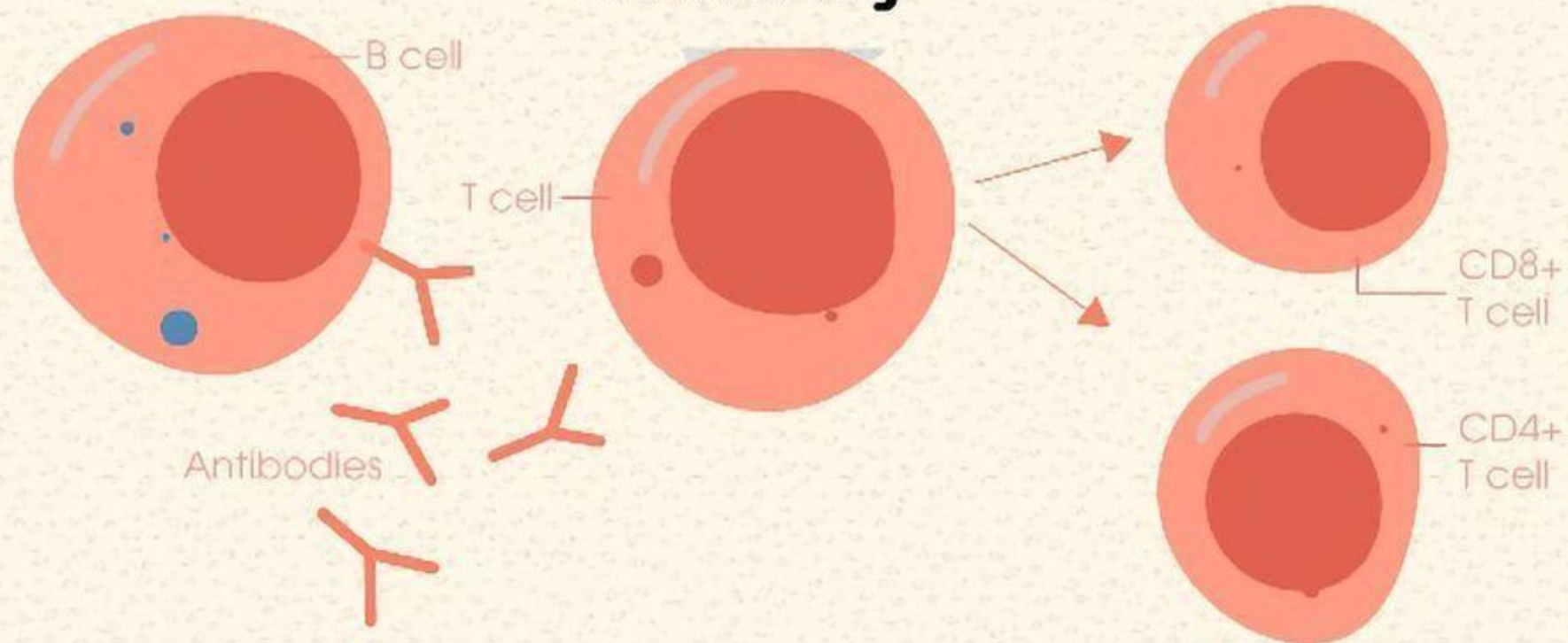
It is the most powerful immune mechanism that protects the body from invading organisms or toxic substances.

Lymphocytes are responsible for the development of acquired immunity.

TYPES OF ACQUIRED IMMUNITY

1. Cell-mediated immunity.
2. Humoral immunity.

Humoral **VS** Cellular Immunity



LYMPHOCYTES

In fetus, lymphocytes develop from the bone marrow. All lymphocytes are released in the circulation & are differentiated into two categories:

1. **T lymphocytes:** responsible for the development of cellular immunity.
2. **B lymphocytes:** responsible for humoral immunity.

PREPROCESSING OF T LYMPHOCYTES

T lymphocytes are processed in the thymus. The processing of T lymphocytes occurs mostly during the period between just before birth & few months after birth. Thymus secretes a hormone called **thymosin** which plays an important role in immunity. T lymphocytes after originating in the bone marrow first migrate to the thymus gland where they divide rapidly and develop extreme diversity for reacting against different specific antigens. One lymphocyte develops reactivity against one specific antigen and the next lymphocyte develops specificity against another antigen.

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The process continues until there are thousands of different types of lymphocytes with specific reactivities against many thousands of different antigens. These different types of preprocessed T lymphocytes now leave the thymus and spread via blood throughout the body to lodge in lymphoid tissue everywhere.

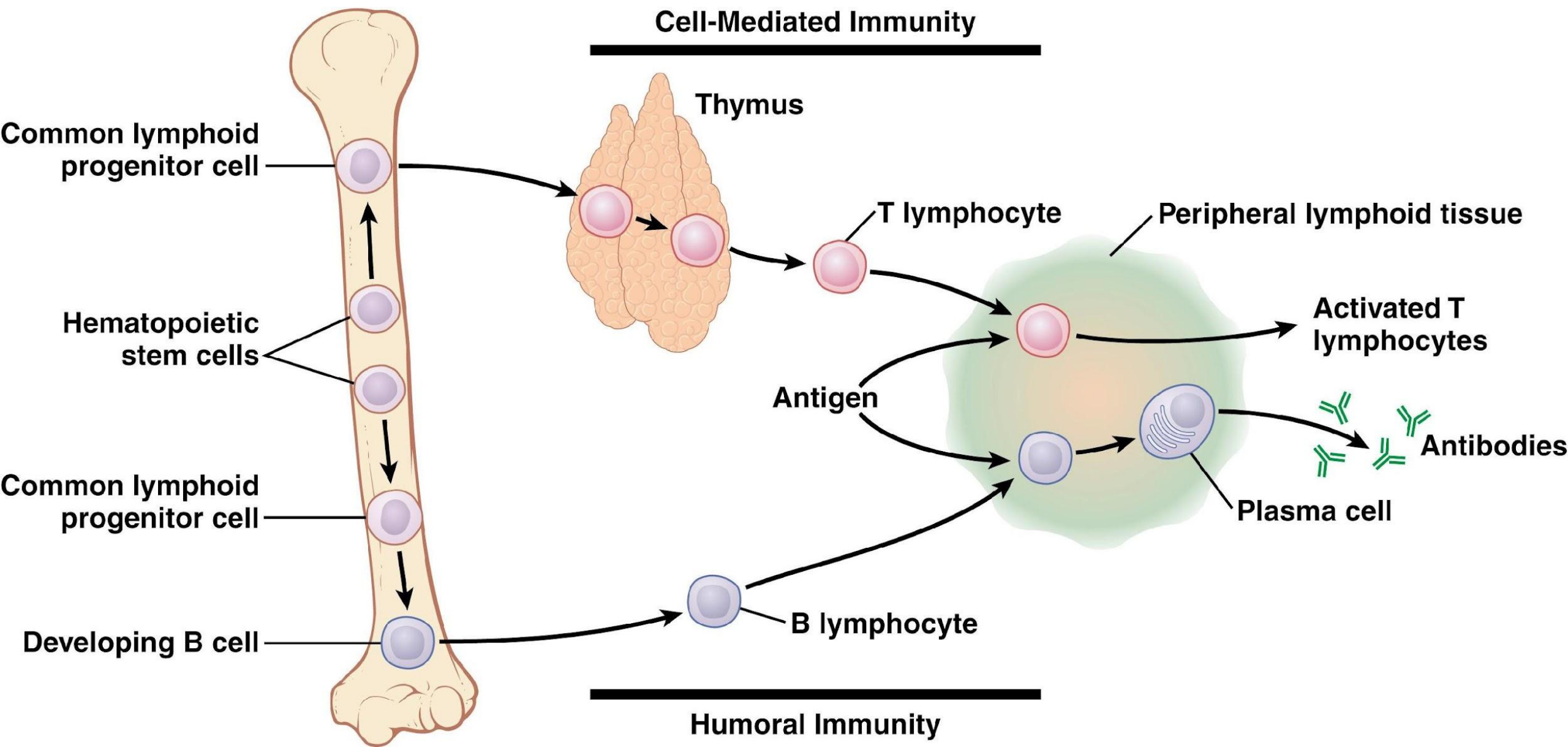
The thymus also makes sure that any T lymphocyte leaving the thymus will not react against proteins or other antigens present in the body's own tissues. Thymus selects which T lymphocytes will be released by first mixing them with virtually all the specific self antigens. If a T lymphocyte reacts, it is destroyed and phagocytized instead of being released. Thus, cells that are finally released are those that are non-reactive against body's own self antigens & they react only against antigens from outside source such as a bacterium, toxin or even transplanted tissue from another person.

PREPROCESSING OF B-LYMPHOCYTES

In humans, B lymphocytes are preprocessed in the liver during midfetal life & in the bone marrow during late fetal life & after birth. B lymphocytes are different from T lymphocytes in two ways.

1. Instead of whole cell developing reactivity against the antigen, the B lymphocytes actively secrete antibodies that are the reactive agents. These antibodies are large proteins capable of combining with & destroying the antigenic substance.
2. The B lymphocytes have even greater diversity than the T lymphocytes thus forming many millions types of antibodies with different specific reactivities.

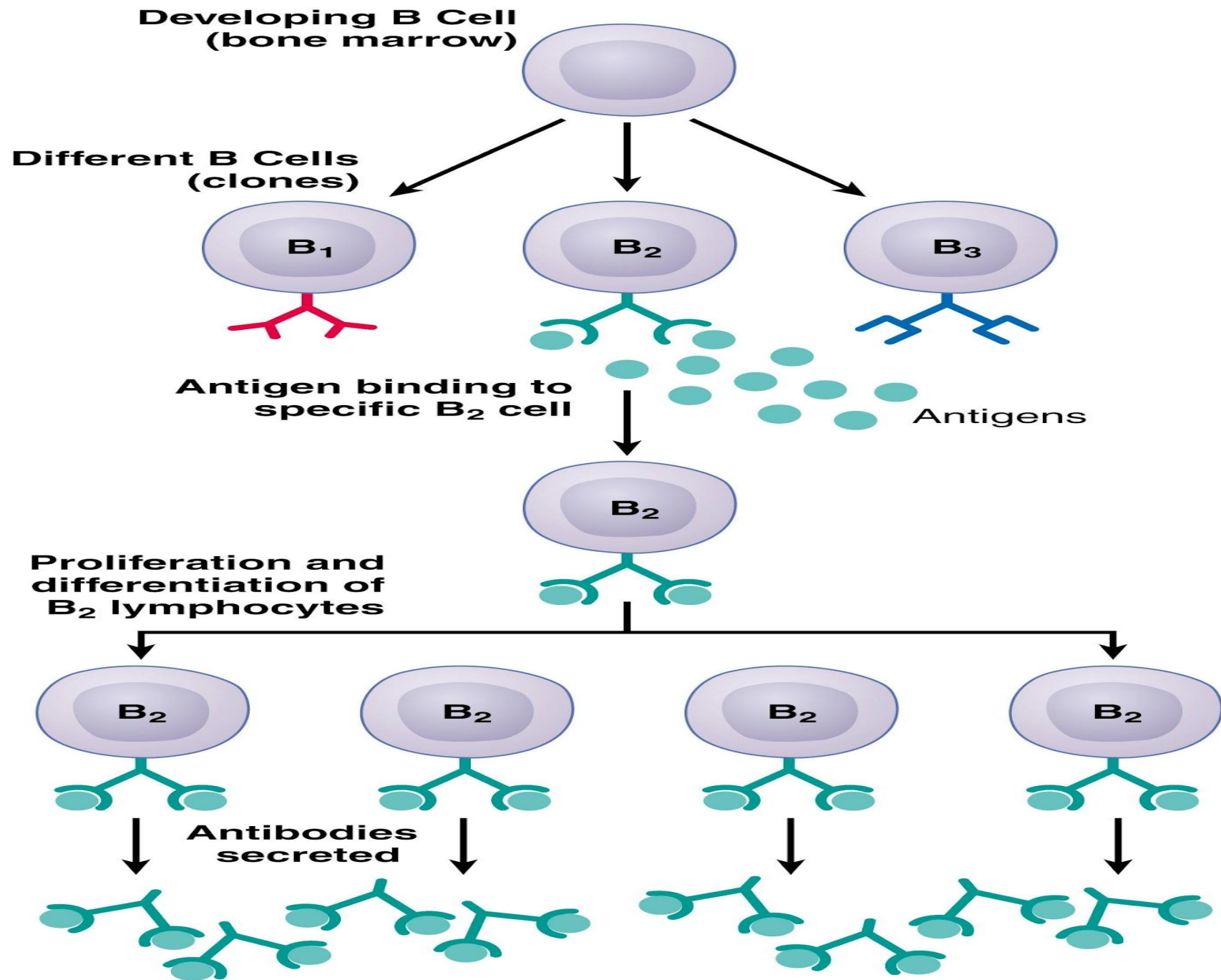
After preprocessing the B lymphocytes also migrate to the lymphoid tissue throughout the body.



LYMPHOCYTE CLONE

When specific antigen comes into contact with T & B lymphocytes in the lymphoid tissue, some of the T lymphocytes become activated to form activated T cells & some of the B lymphocytes become activated to form antibodies. The activated T cells and antibodies now react highly specifically against the particular types of antigens that initiated their development.

All the different lymphocytes capable of forming one specific antibody or T cell are called a clone of lymphocytes.



ANTIGENS

Antigens are the substances which induce specific immune reactions in the body.

Antigens are mostly the conjugated proteins like lipoproteins, glycoproteins & nucleoproteins.

Antigens are of two types:

- 1. Auto antigens or self antigens:** present on the body's own cells.
- 2. Foreign antigens or non-self antigens:** that enter the body from outside.
 - Receptors on the cell membrane of microbial organisms such as bacteria, virus & fungi.
 - Toxins from microbial organisms.
 - Materials from transplanted organs or incompatible blood cells.
 - Allergens or allergic substances like pollen grains.

CELL MEDIATED IMMUNITY

Cell mediated immunity also called *cellular immunity* or *T cell immunity* is the major defense mechanism against infections by viruses, fungi & few bacteria like tubercle bacillus. It is also responsible for delayed allergic reactions and the rejection of transplanted tissues.

Cell mediated immunity is offered by T lymphocytes and it starts developing when T cells come in contact with the antigens. Usually the invading organisms carry the antigenic materials. These antigenic materials are released from invading organisms & are presented to the helper T cells by antigen presenting cells.

ANTIGEN PRESENTING CELLS

Antigen presenting cells are of three types:

1. Macrophages
2. Dendritic cells
3. B lymphocytes

MACROPHAGES

Macrophages are the major antigen presenting cells. These are large phagocytic cells which digest the invading organisms to release the antigen. Macrophages are present along with lymphocytes in almost all the lymphoid tissues.

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DENDRITIC CELLS

Dendritic cells are the most potent antigen presenting cells. Dendritic cells are non-phagocytic in nature and classified into three categories based on their location.

1. Dendritic cells of spleen which trap the antigen in blood.
2. Follicular dendritic cells in lymph nodes which trap the antigen present in lymph.
3. Langerhans dendritic cells in skin which trap organisms coming in contact with body surface.

B LYMPHOCYTES

B lymphocytes also act as antigen presenting cells. Thus the B cells function as both antigen presenting cells & antigen receiving cells. However, B cells are the least efficient antigen presenting cells & need to be activated by helper T cells.

MAJOR HISTOCOMPATIBILITY COMPLEX(MHC) PROTEINS & ANTIGEN RECEPTORS ON T LYMPHOCYTES

Unlike B lymphocytes which recognize intact antigen, T lymphocytes respond to antigens only when they are bound to specific molecules called MHC proteins present on the surface of antigen presenting cells. The MHC proteins bind peptide fragments of antigen proteins that are degraded inside the antigen presenting cells & then transport them to the cell surface. There are two types of MHC proteins

1. ***MHC 1 PROTEINS*** which present antigens to cytotoxic T cells.
2. ***MHC 2 PROTEINS*** which present antigens to T helper cells.

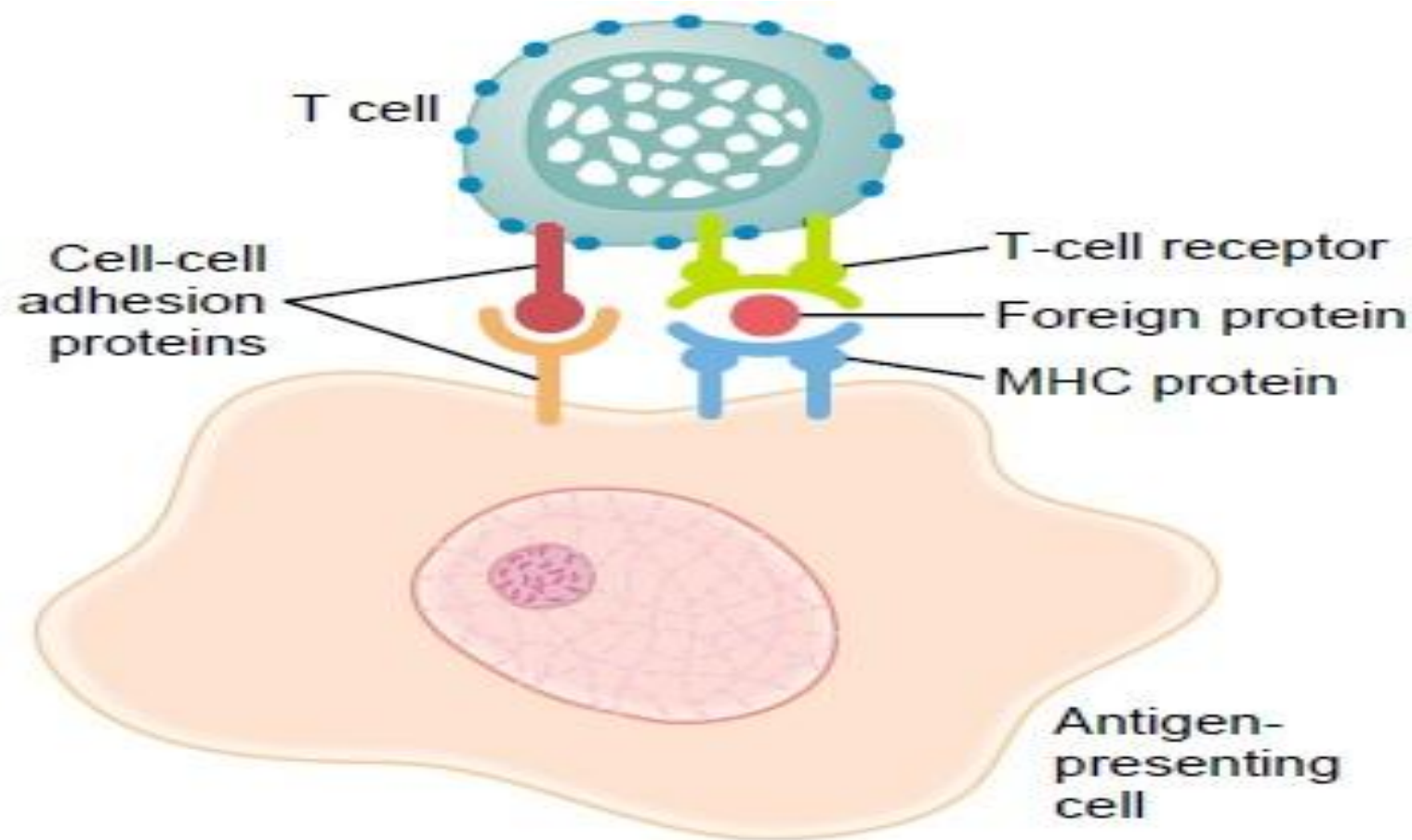


Figure 34–6

Activation of T cells requires interaction of T-cell receptors with an antigen (foreign protein) that is transported to the surface of the antigen-presenting cell by a major histocompatibility complex (MHC) protein. Cell-to-cell adhesion proteins enable the T cell to bind to the antigen-presenting cell long enough to become activated.

HELPER T CELLS

Helper T cells (CD4 cells) activates all the other T cells and B cells.

Helper T cells are of two types:

- Helper-1 T (TH1) cells.
- Helper-2 T (TH2) cells.

ROLE OF TH1 CELLS

The TH1 cells are concerned with cellular immunity & secrete two substances:

1. Interleukin-2 which activates the other T cells.
2. Gamma interferon which stimulates the phagocytic activity of cytotoxic cells, macrophages & natural killer cells.

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ROLE OF TH2 CELLS

The TH2 cells are concerned with humoral immunity & secrete interleukin-4 & interleukin-5 which are concerned with the activation of B cells, proliferation of plasma cells & production of antibodies by plasma cells.

ROLE OF CYTOTOXIC T CELLS IN CELL MEDIATED IMMUNITY

Cytotoxic T cells activated by helper T cells circulate through blood, lymph & lymphatic tissues and destroy the invading organism by attacking them directly.

MECHANISM OF ACTION OF CYTOTOXIC T CELLS

1. Receptors present on the outer membrane of cytotoxic T cells bind the antigens or organisms tightly with cytotoxic T cells.
2. Cytotoxic T cells enlarge & release cytotoxic substances like lysosomal enzymes.
3. In this way each cytotoxic T cell can destroy a large no. of microorganisms one after another.

Cytotoxic T cells also destroy cancer cells, transplanted cells like those of transplanted heart, kidney etc which are foreign bodies.

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Cytotoxic T cells also destroy even the bodies's own tissues which are affected by foreign bodies particularly the viruses. Many viruses are entrapped in the membrane of affected cells. The antigen of the viruses attracts T cells & the cytotoxic T cells kill the affected cells along with the viruses. Due to this the cytotoxic T cells are also called **KILLER CELLS**.

SUPPRESSOR T CELLS

Suppressor T cells also called regulatory T cells suppress the activities of killer T cells. So, the suppressor T cells play an important role by preventing the killer T cells from destroying body's own tissues along with invaded organisms. These cells suppress the activities of helper T cells also.

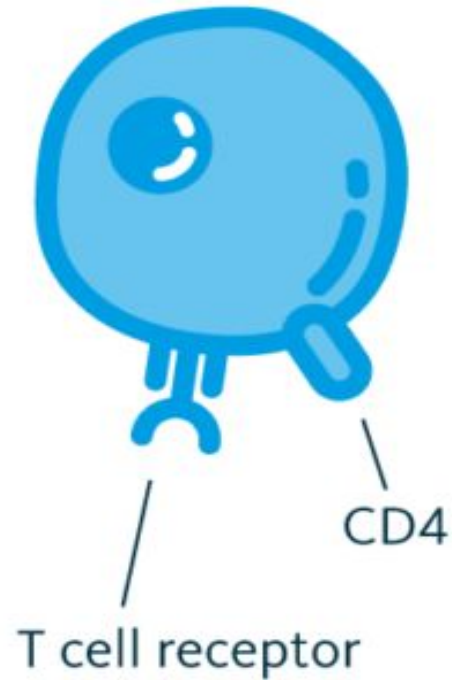
MEMORY T CELLS

Some of the T cells activated by an antigen do not enter the circulation and remain in the lymphoid tissue. These T cells are called *memory T cells*.

In later periods, these memory T cells migrate to various lymphoid tissues throughout the body & when the body is exposed to the same organism for the next time, these memory cells identify the organism and immediately activate the other T cells. So the invading organism is destroyed very quickly & the response of T cells is also more powerful this time.

Different types of T cells

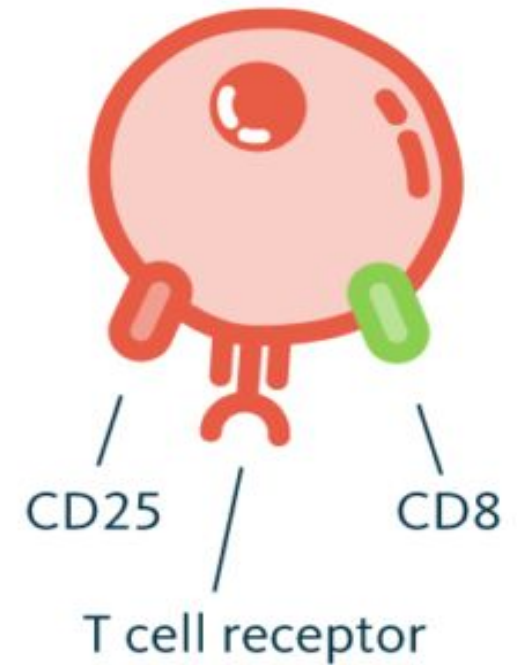
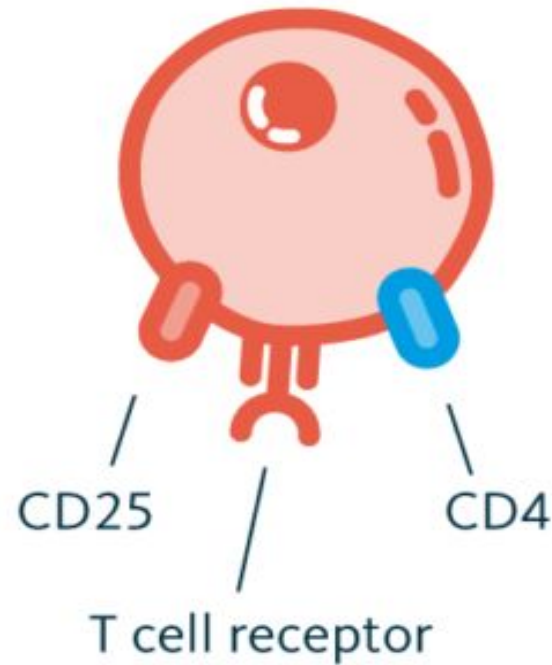
T Helper cell



Cytotoxic T cell



T Regulatory cells



SPECIFICITY OF T CELLS

Each T cell is designed to be activated by only one type of antigen. It is capable of developing immunity against that particular antigen only. This property is called the specificity of T cells.

HUMORAL IMMUNITY

Humoral immunity is defined as the immunity mediated by antibodies which are secreted by B lymphocytes. B lymphocytes secrete antibodies into the body fluids (blood & lymph) called humors in latin. Since B lymphocytes provide immunity through humors, it is called as humoral immunity or B cell immunity. The humoral immunity is the major defense mechanism against bacterial infections.

Antibodies are the gamma globulins produced by B lymphocytes. The antibodies fight against the invading organisms in two ways:

- 1. By direct attack on the invader &**
- 2. By activation of the complement system.**

ANTIBODIES (IMMUNOGLOBULINS)

An antibody is defined as:

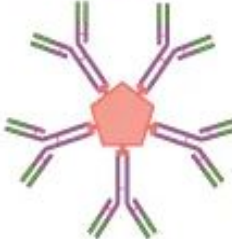
a protein that is produced by B lymphocytes in response to the presence of an antigen.

Antibodies are gamma globulins & form 20% of total plasma proteins. Antibodies enter almost all the tissues of the body.

TYPES OF ANTIBODIES

1. IgA
2. IgD
3. IgE
4. IgG
5. IgM

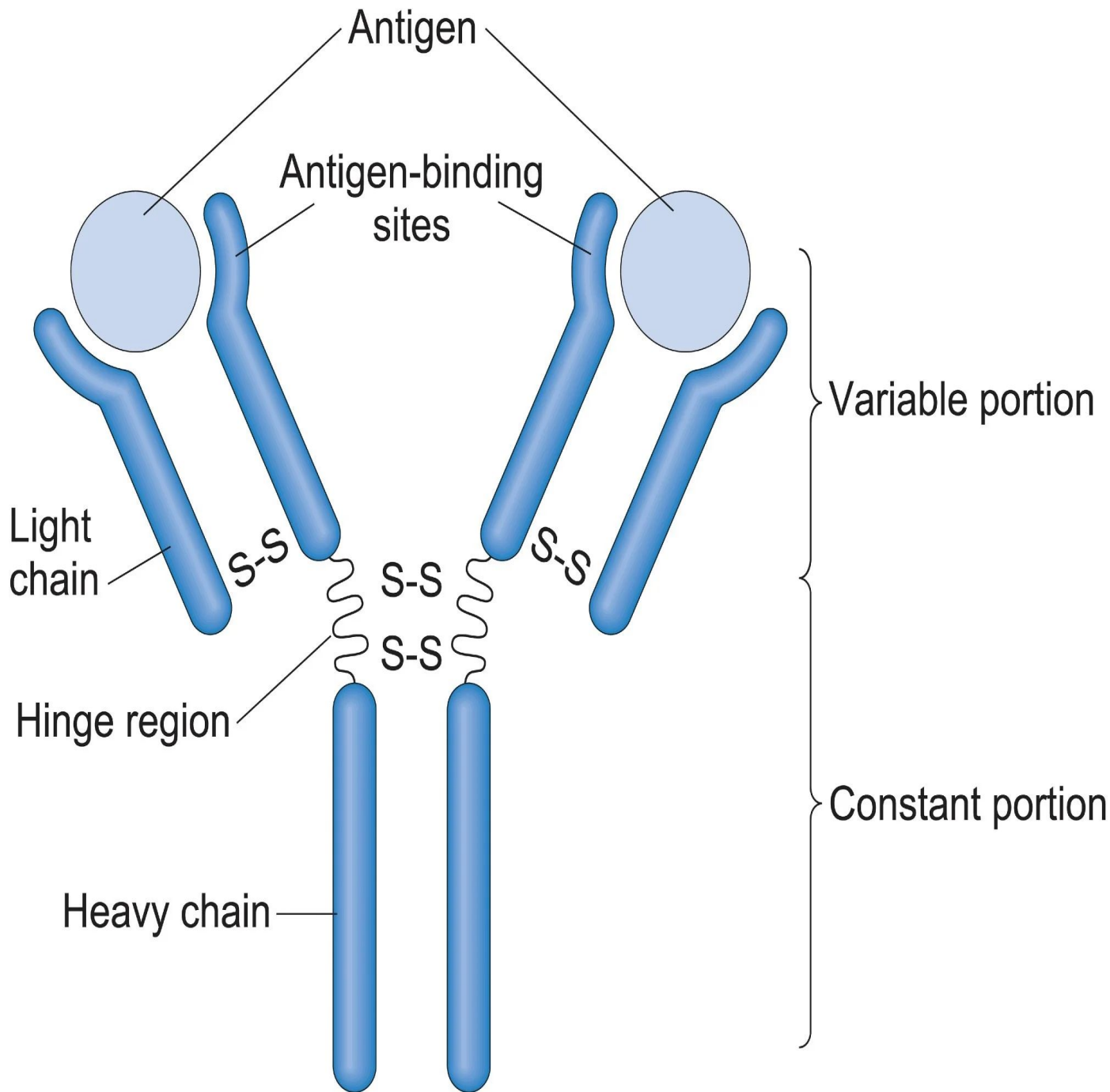
Among these antibodies IgG forms 75% of total antibodies in the body.

Name	Properties	Structure
IgA	Found in mucous, saliva, tears, and breast milk. Protects against pathogens.	
IgD	Part of the B cell receptor. Activates basophils and mast cells.	
IgE	Protects against parasitic worms. Responsible for allergic reactions.	
IgG	Secreted by plasma cells in the blood. Able to cross the placenta into the fetus.	
IgM	May be attached to the surface of a B cell or secreted into the blood. Responsible for early stages of immunity.	

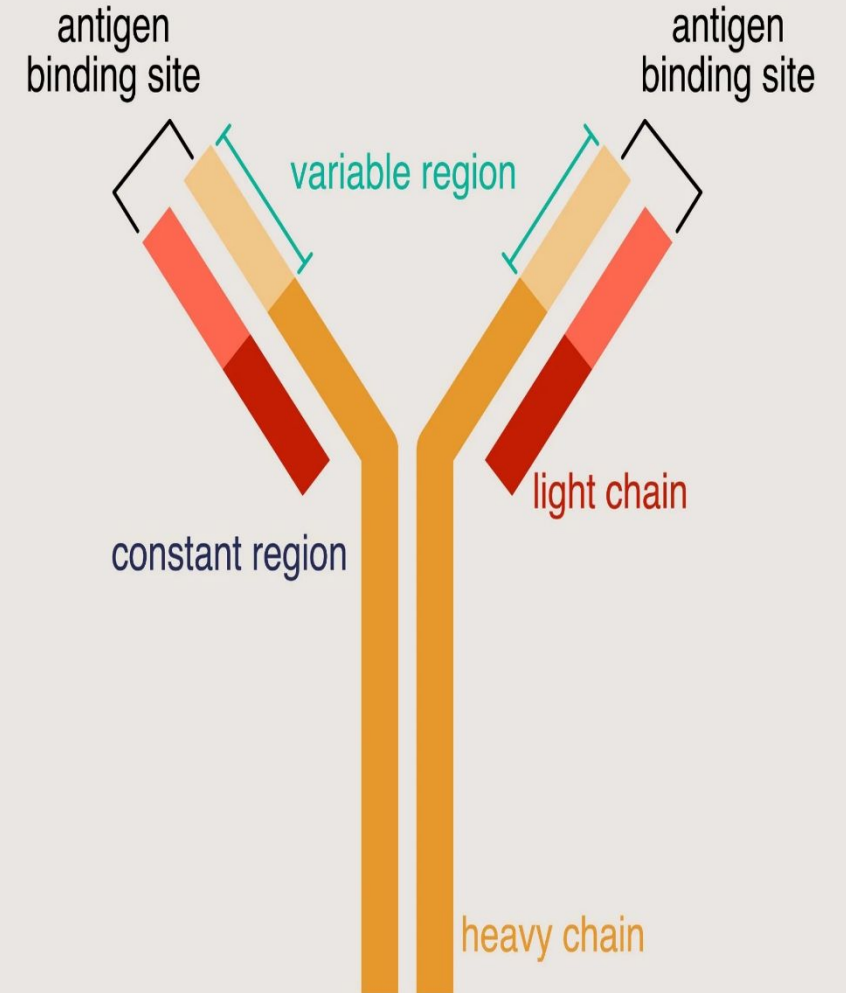
STRUCTURE OF ANTIBODIES

Antibodies have a molecular weight of **150,000 to 900,000**. The antibodies are formed by two pairs of chains namely one pair of heavy or long chains & one pair of light or short chains. Each heavy chain consists of about 400 amino acids & each light chain consists of about 200 amino acids.

Actually each antibody has two halves which are identical. The two halves are held together by **disulfide bonds(S-S)**. Each half of the antibody consists of one **heavy chain(H)** & one **light chain(L)**. The two chains in each half are also joined by disulfide bonds(S-S). The disulfide bonds allow the movement of amino acid chains. In each antibody the light chain is parallel to one end of the heavy chain. The light chain and the part of heavy chain parallel to it form one arm. The remaining part of the heavy chain forms another arm. A **hinge** joins both arms. Each chain of antibody includes two regions (constant region & variable region).



ANTIBODY STRUCTURE



CONSTANT REGION

Amino acids present in this region are similar in number & placement (sequence) in all antibodies of each type. So this region is called **constant region** or **Fc (fragment crystallizable) region**. Thus the identification & functions of different types of immunoglobulins depend upon the constant region.

This region causes complement fixation so this region is also called the **complement binding region**.

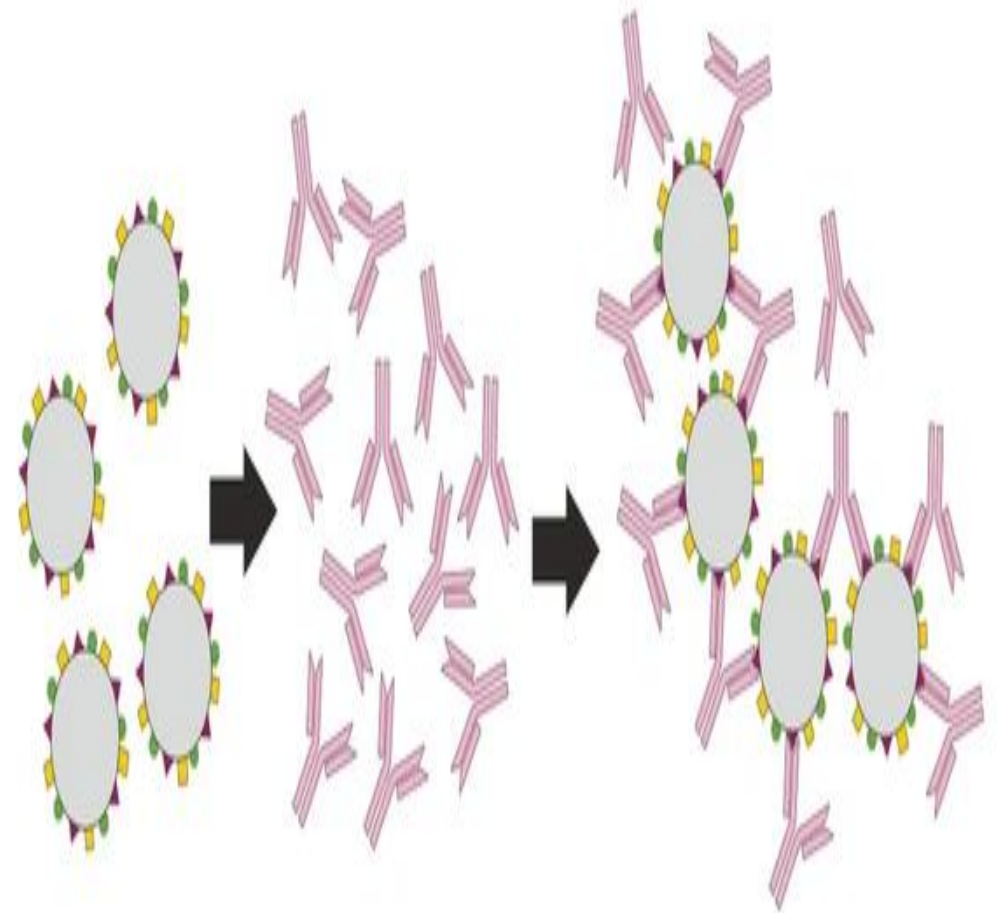
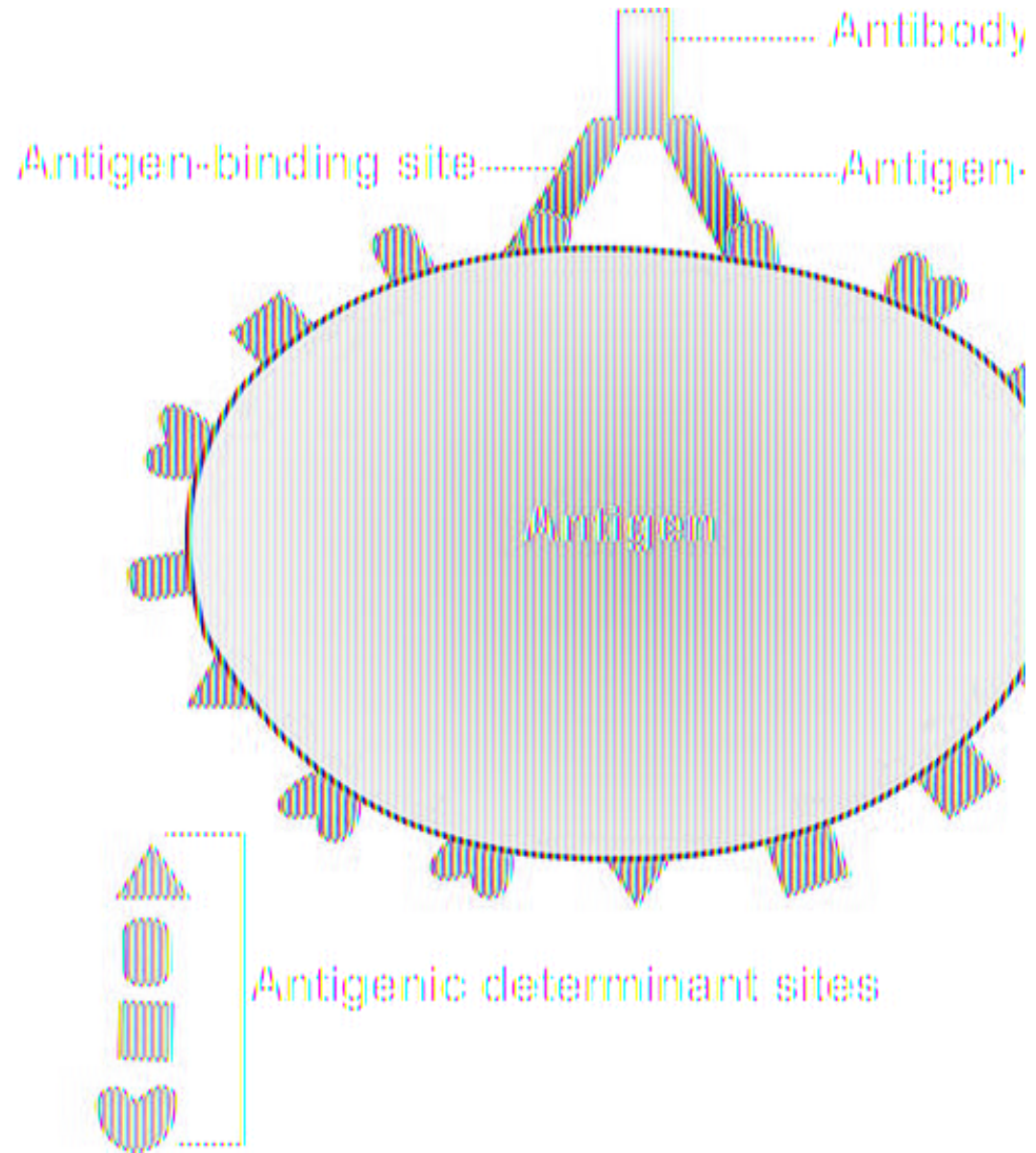
VARIABLE REGION

Variable region is smaller as compared to the constant region. Amino acids in this region are different in number & placement(sequence) in each antibody. So it is called as variable region. This region enables the antibody to recognize the specific antigen & to bind itself with the antigen. So this region of chain is called **antigen binding region** or **Fab(fragment antigen binding) region**.

DIRECT ACTION OF ANTIBODIES ON INVADING ORGANISM

The antibodies can inactivate the invading organism directly in one of the following ways:

1. **AGGLUTINATION:** in which multiple large particles with antigens on their surface are bound together into a clump.
2. **PRECIPITATION:** in which the antigen-antibody complex becomes so large that it is rendered insoluble & precipitates.
3. **NEUTRALIZATION:** in which the antibodies cover the toxic sites of the invading organism.
4. **LYSIS:** in which some potent antibodies are occasionally capable of directly attacking membranes of cellular agents and cause rupture of the agent.



COMPLIMENT SYSTEM OF ANTIBODY ACTION

It is a system of about 20 proteins present normally among the plasma proteins in the blood & also among the proteins that leak out of capillaries into the tissue spaces.

The direct actions of antibodies are usually not strong enough to play a major role in protecting the body against the invader. So most of the protection occurs through the amplifying effects of the compliment system.

The main function of the compliment system is to enhance or accelerates various activities during the fight against the invading organism.

The proteins of compliment system are present in inactive form & are activated in three ways:

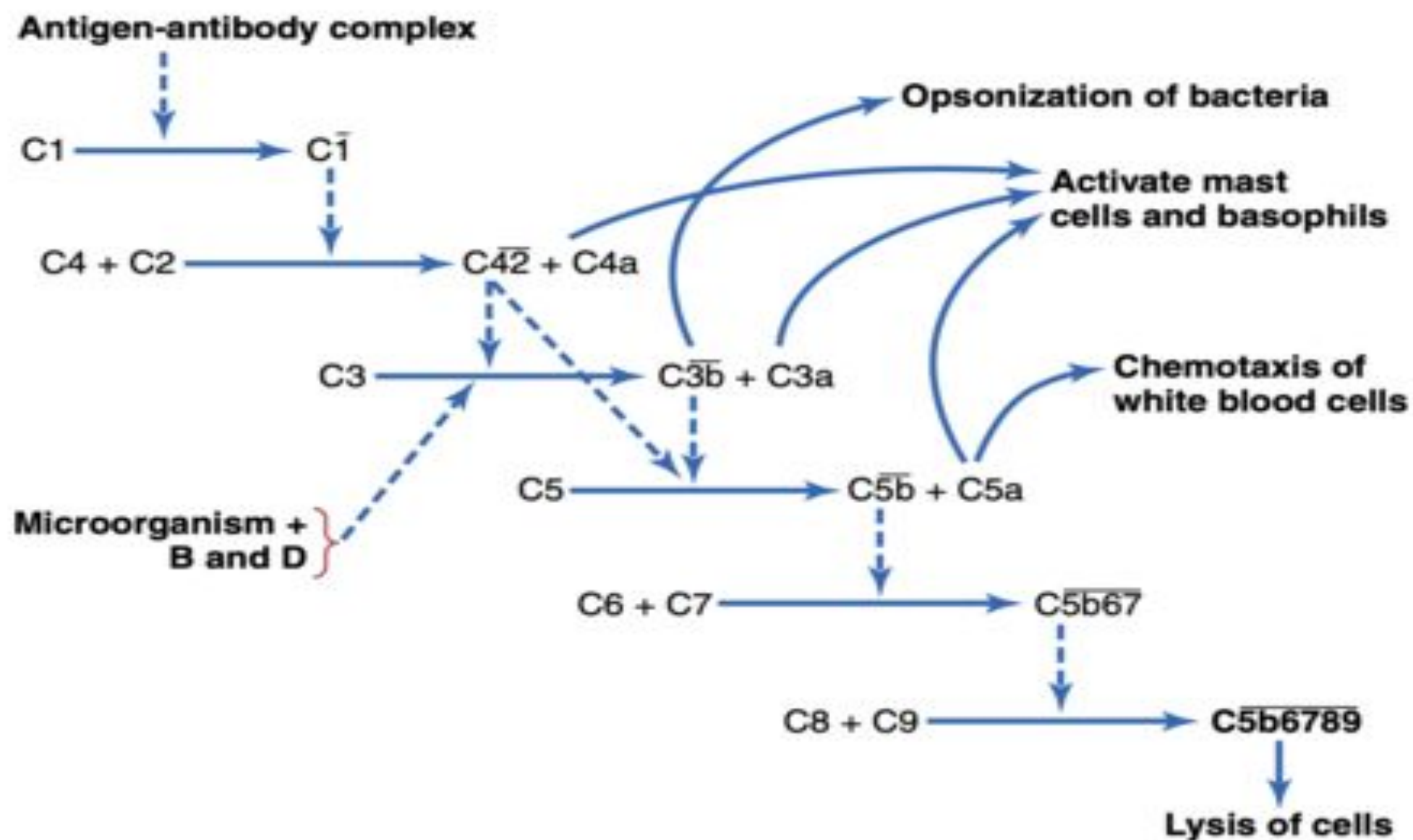


Figure 35-6. Cascade of reactions during activation of the classical pathway of complement.

1. CLASSICAL PATHWAY

The classical pathway is initiated by an antigen-antibody reaction. In classical pathway, C1 binds with the antigen-antibody complex & triggers a series of events in which other enzymes are activated in sequence. These enzymes or the by-products formed during these events produce the following activities:

OPSONIZATION: one of the products of compliment cascade, C3b strongly activates phagocytosis by neutrophils & macrophages, causing these cells to engulf the bacteria. This process is called opsonization.

LYSIS: One of the most important of all the products of compliment cascade called the **membrane attack complex (cytolytic complex)** inserts itself into the lipid bilayer of the cell membrane creating pores in the cell membrane of bacteria causing osmotic rupture of the cell membranes of bacteria or other invading organism.

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AGGLUTINATION: The compliment products also change the surfaces of invading organisms causing them to adhere to one another thus promoting agglutination.

NEUTRALIZATION: The compliment enzymes & other compliment products can attack the structures of some viruses & thereby render them nonvirulent.

CHEMOTAXIS: Fragment C5a initiates chemotaxis of neutrophils & macrophages thus causing large numbers of these phagocytes to migrate into the tissue area adjacent to the antigenic agent.

ACTIVATION OF MAST CELLS AND BASOPHILS: Fragments C3a, C4a & C5a activate the mast cells & basophils which release histamine. Histamine dilates the blood vessels & increases capillary permeability so plasma proteins from blood enter the tissues & inactivate the antigenic products.

CONT..

INFLAMMATORY EFFECTS: In addition to the inflammatory effects caused by the activation of mast cells & basophils, several other complement products contribute to local inflammation. These products cause the already increased blood flow to increase still further, the capillary leakage of proteins to be increased & the interstitial fluid proteins to coagulate in the tissue spaces thus preventing movement of the invading organism through the tissues.

2. LECTIN PATHWAY

Lectin pathway occurs when mannose binding lectin (MBL) which is a serum protein binds with mannose or fructose group on wall of bacteria, fungi or virus.

3. ALTERNATE PATHWAY

Compliment system is also activated by another way called alternate pathway. It is due to a protein in circulation called factor 1. It binds with polysaccharides present in the cell membrane of the invading organisms. This binding activates C3 & C5 which ultimately attack the antigenic products of invading organism.

PASSIVE IMMUNIZATION

Passive immunization(immunity) is produced without challenging the immune system of the body.

It is achieved by the administration of serum or antibodies from a person who is already immunized(affected) by a disease to a non-immune person.

Passive immunization can be achieved in two ways:

1. Passive natural immunization
2. Passive artificial immunization

PASSIVE NATURAL IMMUNIZATION

- Acquired from mother before birth from maternal antibodies(mainly IgG) through placenta.
- Acquired from mother after birth through IgA antibodies present in breast milk.
- Short period immunity because of metabolization of mother antibodies.

PASSIVE ARTIFICIAL IMMUNIZATION

- Obtained by injecting previously prepared antibodies using serum from humans or animals.
- Antibodies are obtained from the person affected by the disease or from animals immunized artificially.
- Used prophylactically against infections like tetanus, measles, diphtheria etc & also for snake & insects poisoning.

POST ASSESSMENT

Q1. Two major types of immunity are:

- a) Cellular & hormonal
- b) Innate & adaptive
- c) Active & cellular
- d) Passive & innate

Q2. Which of the following is a characteristic of innate immunity:

- a) Specific response
- b) Immunologic memory
- c) Antibody production
- d) Immediate defense

CONT..

Q3. Adaptive immunity is characterized by:

- a) Lack of specificity
- b) Rapid but temporary response
- c) Specificity & memory
- d) Absence of lymphocytes

Q4. Antibodies are produced by:

- a) Plasma cells
- b) Macrophages
- c) Neutrophils
- d) T lymphocytes



THANK YOU!